

CBSE Class 12 Physics — Magnetism and Matter

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

- Q1.** The magnetic susceptibility of a magnetic material is -4.2×10^{-6} . Identify the nature of the magnetic material. [PYQ 2022] Very High
- Q2.** A magnetic dipole moment of a bar magnet is a vector quantity directed: (a) upward at perpendicular bisector (b) from North pole to South pole (c) downward at perpendicular bisector (d) from South pole to North pole. [PYQ 2022] Very High
- Q3.** If the magnetizing field (H) applied to a ferromagnetic material is steadily increased, its magnetic permeability (μ): (a) decreases (b) increases (c) remains unchanged (d) first decreases and then increases. [PYQ 2022] High
- Q4.** Write the mathematical form of Gauss's Law for magnetism. What does it physically signify regarding magnetic monopoles? [PYQ 2020] Very High
- Q5.** What is the amount of work done in rotating a magnetic dipole of moment m from stable equilibrium to unstable equilibrium in a uniform magnetic field B ? [Practice] High
- Q6.** For a given magnetic material, the relative permeability is $\mu_r = 0.998$. Is the material diamagnetic, paramagnetic, or ferromagnetic? [PYQ 2020] Very High
- Q7.** Which of the following is the most suitable material for the core of an electromagnet? (a) Steel (b) Soft iron (c) Copper (d) Aluminum. [PYQ 2021] Very High
- Q8.** A permanent magnet should have: (a) high retentivity and high coercivity (b) low retentivity and low coercivity (c) high retentivity and low coercivity (d) low retentivity and high coercivity. [PYQ 2021] Very High
- Q9.** State the dimensional formula and the SI unit of magnetic susceptibility (χ). [PYQ 2019] Medium
- Q10.** According to Curie's law, how does the magnetic susceptibility (χ) of a paramagnetic substance depend on its absolute temperature (T)? [Practice] High
- Q11.** Which of the following magnetic materials is strictly repelled by an external magnetic field: Diamagnetic, Paramagnetic, or Ferromagnetic? [PYQ 2020] Very High
- Q12.** What happens to the magnetic properties of a ferromagnetic substance when it is heated significantly above its Curie temperature? [PYQ 2021] High
- Q13.** Two magnetic materials A and B have relative magnetic permeabilities of 0.96 and 500 respectively. Identify the magnetic nature of materials A and B. [PYQ 2018] Very High
- Q14. Assertion (A):** Magnetic field lines form continuous closed loops. **Reason (R):** Isolated magnetic monopoles do not exist in nature. [PYQ 2022] Very High

Q15. Assertion (A): Diamagnetic materials are feebly repelled by magnets. **Reason (R):** They develop an induced magnetic dipole moment in a direction opposite to the applied magnetic field. [Practice] High

Q16. Assertion (A): Soft iron is preferred over steel for making the core of a transformer. **Reason (R):** The area of the hysteresis loop (B-H curve) for soft iron is much smaller than that for steel, resulting in lower energy loss. [Practice] Very High

Q17. A small piece of material tends to move from the stronger to the weaker part of a non-uniform magnetic field. Identify the class of magnetic material. [PYQ 2016] High

Q18. Write the SI unit of magnetic dipole moment. [Practice] High

Q19. An electron revolving around a nucleus possesses a magnetic dipole moment. If its orbital angular momentum is L , what is the magnitude of its magnetic moment? [PYQ 2023] Very High

Q20. Does the magnetic susceptibility of a diamagnetic substance depend upon temperature? [Practice] Medium

2-Mark Questions

Q21. State Gauss's law for magnetism. Explain its core physical significance. [PYQ 2023] Very High

Q22. Write four important properties of magnetic field lines due to a bar magnet. [PYQ 2023] Very High

Q23. Define the terms 'magnetization' (M) and 'magnetic intensity' (H). State their SI units. [Practice] High

Q24. A magnetic material has a relative permeability $\mu_r = 0.5$. Identify the nature of the magnetic material and write its mathematical relation to magnetic susceptibility (χ). [PYQ 2020] Very High

Q25. A magnetic dipole of moment m is aligned parallel to an external magnetic field B . Work of 0.25 J has to be done to turn it through an angle of 60° . Find the external counter torque required to perfectly maintain the dipole at this angle. [PYQ 2022] High

Q26. Distinguish clearly between a diamagnetic substance and a paramagnetic substance in respect to their behavior in a uniform and a non-uniform magnetic field. [PYQ 2016] Very High

Q27. A uniform magnetic field gets modified (field lines are pushed out of the material vs field lines crowd into the material) when two different specimens (i) and (ii) are placed in it. Identify the magnetic nature of specimen (i) and (ii), and compare their magnetic susceptibilities. [PYQ 2022] High

Q28. Briefly explain the origin of diamagnetism on the basis of the orbital motion of paired electrons in atoms. [Practice] Medium

- Q29.** What happens to the magnetic dipole moment of a bar magnet if it is cut into two perfectly equal pieces (i) transverse to its length, and (ii) along its length? [Practice] High
- Q30.** Write two specific properties of a material suitable for making (a) a permanent magnet, and (b) an electromagnet. [PYQ 2017] Very High
- Q31.** A closely wound solenoid of 2000 turns and cross-sectional area $1.6 \times 10^{-4} \text{ m}^2$, carrying a current of 4.0 A , is suspended through its center, allowing it to turn in a horizontal plane. Calculate the magnetic dipole moment associated with this solenoid. [PYQ 2022] High
- Q32.** Deduce the relationship between relative magnetic permeability (μ_r) and magnetic susceptibility (χ). [Practice] Medium
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3-Mark Questions

- Q33.** A bar magnet of magnetic moment 6 J/T is aligned at 60° with a uniform external magnetic field of 0.44 T . Calculate: (i) the work done in turning the magnet to align its magnetic moment normal to the magnetic field, (ii) the work done to align it opposite to the magnetic field, and (iii) the torque on the magnet in the final orientation of case (ii). [PYQ 2023] Very High
- Q34.** Define the term magnetic susceptibility. Briefly explain the temperature dependence of susceptibility for diamagnetism, paramagnetism, and ferromagnetism from a molecular/atomic viewpoint. [PYQ 2019] Very High
- Q35.** A small compass needle of magnetic moment m and moment of inertia I is free to oscillate in a uniform magnetic field B . It is slightly disturbed from its equilibrium position and then released. Show that it executes simple harmonic motion, and write the expression for its time period. [PYQ 2023] High
- Q36.** Derive the expression for the potential energy of a magnetic dipole of moment $\vec{m}$ placed in a uniform magnetic field $\vec{B}$. [Practice] High
- Q37.** Two identical magnetic materials, P and Q, are subjected to a magnetizing field. The B-H curve for P is narrow, while for Q it is broad. (a) What does the area under the B-H curve represent? (b) Which of the two materials is more suitable for making the core of a transformer, and why? [Practice] Very High
- Q38.** Draw the magnetic field lines when a diamagnetic and a paramagnetic substance are placed in a uniform external magnetic field. Which fundamental magnetic property distinguishes their behavior in the field? [PYQ 2021] High
- Q39.** Define the terms 'Coercivity' and 'Retentivity' for a ferromagnetic material. Between soft iron and steel, which has higher retentivity and which has higher coercivity? [PYQ 2017] Very High
- Q40.** A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.25 T experiences a torque of magnitude $4.5 \times 10^{-2} \text{ J}$. What is the magnitude of the

magnetic moment of the magnet? Calculate the potential energy of the magnet in its stable equilibrium position. [PYQ 2018] High

Q41. Using the qualitative model of a bar magnet as an equivalent solenoid, write the expression for the magnetic field produced by a short bar magnet at a distance r on its (i) axial line, and (ii) equatorial line. How do these fields depend on the distance r ? [Practice] High

Q42. The relative magnetic permeability of a substance is 5000. Identify the substance. If this substance is heated beyond a critical temperature, what change occurs in its magnetic properties? Name this critical temperature. [Practice] Medium

5-Mark Questions (Long Answer/Case-Study)

Q43. Case Study: Magnetic Properties of Materials Materials are classified into diamagnetic, paramagnetic, and ferromagnetic based on their response to an external magnetic field. Diamagnetic materials possess no net permanent magnetic moment at the atomic level, whereas paramagnetic and ferromagnetic materials have permanent atomic dipoles. Ferromagnetic materials exhibit a strong ordering of these dipoles into regions called domains. (a) What is the net magnetic moment of an individual atom of a diamagnetic material? (b) What physically happens to the domains of a ferromagnetic material when a strong external magnetic field is applied? (c) State the exact mathematical relation between relative permeability (μ_r) and magnetic susceptibility (χ). (d) Why does the susceptibility of a paramagnetic material strongly depend on temperature, while that of a diamagnetic material does not? [Practice] Very High

Q44. Case Study: Hysteresis Loop and Material Selection The B-H curve (hysteresis loop) illustrates how the magnetic field (B) inside a ferromagnetic material varies with the applied magnetizing field (H). The curve demonstrates that the material retains some magnetism even when H is reduced to zero, and a reverse field must be applied to demagnetize it entirely. (a) What physical quantity does the total area under the B-H hysteresis loop represent? (b) For manufacturing a permanent magnet, what specific kinds of retentivity and coercivity are required? (c) Why is soft iron vastly preferred over steel for making the cores of transformers and AC generators? (d) Define the terms 'Retentivity' and 'Coercivity'. [PYQ 2020] Very High

Q45. (A) Derive the expression for the torque acting on a magnetic dipole (such as a current loop or bar magnet) of dipole moment $\vec{m}$ placed in a uniform magnetic field $\vec{B}$. (B) A square coil of side **10 cm** consisting of 20 turns carries a current of **12 A**. The coil is suspended vertically, and the normal to its plane makes an angle of **30°** with a uniform horizontal magnetic field of **0.80 T**. Calculate the magnitude of the torque experienced by the coil. (C) In which orientation will the coil be in (i) stable equilibrium, and (ii) unstable equilibrium? [PYQ 2019/2022] High

Q46. (A) Explain the origin of paramagnetism and diamagnetism on the basis of atomic structure and paired/unpaired electrons. (B) How does a ferromagnetic material differ fundamentally from a paramagnetic material in its internal structure and response to an applied magnetic field? (C) A magnetic material has a susceptibility of $\chi = 2.6 \times 10^{-5}$. Identify the exact type of magnetic material and state how its relative permeability compares to 1. [PYQ 2019] Very High

Q47. (A) State Gauss's Law for magnetism. How does its mathematical formulation differ from Gauss's Law for electrostatics, and what is the fundamental reason for this difference? (B) A long solenoid has a core of a material with relative permeability **400**. The windings of the solenoid are completely insulated from the core and carry a steady current of **2 A**. If the number of turns is **1000 per metre**, calculate: (i) the magnetic intensity **H** , (ii) the net magnetic field **B** inside the core, and (iii) the magnetization **M** of the core material. [PYQ 2025-26 Sample Paper] Very High

Q48. (A) Describe qualitatively the behavior of a small magnetic dipole when placed in a *non-uniform* magnetic field. Will it experience a force, a torque, or both? (B) A short bar magnet has a magnetic moment of **0.48 J/T** . Give the direction and magnitude of the magnetic field produced by the magnet at a distance of **10 cm** from the centre of the magnet on (i) the axis, and (ii) the equatorial line of the magnet. (C) Show diagrammatically the magnetic field lines around a typical bar magnet. [Practice] High

Q49. Case Study: Electromagnets and Permanent Magnets Electromagnets are created by placing a highly permeable magnetic core inside a current-carrying solenoid, effectively amplifying the magnetic field. Permanent magnets, however, are made from materials that retain their magnetization for a long time even without an external field. (a) Name the specific magnetic material most commonly used for making the core of electromagnets. (b) What is the exact effect on the magnetic field of placing this iron core inside a current-carrying solenoid? (c) State two practical ways to dramatically increase the magnetic strength of an electromagnet. (d) Can we use steel to make an electromagnet? Give a sound scientific reason. [Practice] High

Q50. (A) Draw the standard B-H (hysteresis) curves for soft iron and steel side by side on the same axes. (B) Based directly on these curves, explicitly compare their retentivity, coercivity, and hysteresis loss per cycle. (C) Which of the two materials is more suitable for making (i) permanent magnets, and (ii) cores of electromagnets? Give proper justification for your choices based on their magnetic properties. [PYQ 2015] Very High

Answer Key

Q1: Diamagnetic material (since χ is negative). **Q2:** (d) from South pole to North pole. **Q3:** (a) decreases (As H increases, B eventually saturates, making $\mu = B/H$ decrease). **Q4:** $\oint \vec{B} \cdot d\vec{S} = 0$. It signifies that isolated magnetic monopoles do not exist. **Q5:** $W = mB(\cos 0^\circ - \cos 180^\circ) = mB(1 - (-1)) = 2mB$. **Q6:** Diamagnetic (since $\mu_r < 1$).

Q7: (b) Soft iron. **Q8:** (a) high retentivity and high coercivity. **Q9:** It is a dimensionless quantity ($[M^0 L^0 T^0]$). It has no SI unit. **Q10:** $\chi \propto 1/T$ (Inversely proportional to absolute temperature). **Q11:** Diamagnetic materials. **Q12:** It loses its ferromagnetic properties and behaves as a paramagnetic substance. **Q13:** Material A ($\mu_r < 1$) is diamagnetic; Material B ($\mu_r \gg 1$) is ferromagnetic. **Q14:** (a) Both A and R are true and R is the correct explanation of A. **Q15:** (a) Both A and R are true and R is the correct explanation of A. **Q16:** (a) Both A and R are true and R is the correct explanation of A. **Q17:** Diamagnetic material. **Q18:** Ampere-meter² ($A \cdot m^2$) or Joule/Tesla (J/T). **Q19:** $M = \frac{e}{2m_e} L$. **Q20:** No, diamagnetism is practically independent of temperature. **Q21:** The net magnetic flux through any closed surface is zero ($\oint \vec{B} \cdot d\vec{S} = 0$). Significance: Magnetic poles always exist in pairs (dipoles); isolated monopoles do not exist. **Q22:** (1) Form continuous closed loops. (2) Emerge from North and merge at South outside, but S to N inside. (3) Never intersect. (4) Degree of closeness indicates field strength. **Q23:** Magnetization (M) is net magnetic moment per unit volume. Intensity (H) is the degree to which an external field can magnetize a material. Both have SI unit A/m. **Q24:** Diamagnetic. Relation: $\mu_r = 1 + \chi \implies \chi = \mu_r - 1 = 0.5 - 1 = -0.5$. **Q25:** $W = mB(1 - \cos 60^\circ) \implies 0.25 = mB(1 - 0.5) \implies mB = 0.5$. Torque $\tau = mB \sin 60^\circ = 0.5 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4} \approx 0.433 \text{ N} \cdot \text{m}$. **Q26:** Uniform field: Dia acquires weak magnetization opposite to field; Para acquires weak magnetization parallel to field. Non-uniform field: Dia moves from stronger to weaker part; Para moves from weaker to stronger part. **Q27:** (i) Diamagnetic (lines expelled), χ is small & negative. (ii) Paramagnetic (lines pulled in), χ is small & positive. **Q28:** Electrons orbiting in atoms act as tiny current loops. An external field alters their speed, inducing a net magnetic moment opposite to the applied field (Lenz's Law), causing diamagnetism. **Q29:** (i) Transverse: m becomes half (length halves, pole strength same). (ii) Along length: m becomes half (length same, pole strength halves). **Q30:** (a) Permanent magnet: High retentivity, high coercivity. (b) Electromagnet: High permeability, low retentivity, low coercivity. **Q31:** $m = NIA = 2000 \times 4.0 \times 1.6 \times 10^{-4} = 1.28 \text{ A} \cdot \text{m}^2$. **Q32:** $B = B_0 + B_m = \mu_0 H + \mu_0 M = \mu_0 (H + \chi H) = \mu_0 H (1 + \chi)$. Since $B = \mu_0 \mu_r H$, comparing gives $\mu_r = 1 + \chi$. **Q33:** (i) $W = mB(\cos 60^\circ - \cos 90^\circ) = 6 \times 0.44 \times 0.5 = 1.32 \text{ J}$. (ii) $W = mB(\cos 60^\circ - \cos 180^\circ) = 2.64 \times (0.5 - (-1)) = 3.96 \text{ J}$. (iii) $\tau = mB \sin 180^\circ = 0$. **Q34:** χ measures how easily a substance gets magnetized (M/H). Dia: Independent of temp. Para: $\chi \propto 1/T$ (Curie's Law, thermal agitation randomizes dipoles). Ferro: Decreases with temp, becomes para above Curie temp. **Q35:** Restoring torque $\tau = -mB \sin \theta \approx -mB\theta$. Also $\tau = I\alpha = I \frac{d^2\theta}{dt^2}$. Thus $\frac{d^2\theta}{dt^2} = -\frac{mB}{I} \theta$. Comparing with SHM $\frac{d^2\theta}{dt^2} = -\omega^2 \theta$, we get $\omega = \sqrt{mB/I} \implies T = 2\pi \sqrt{I/mB}$. **Q36:** Work done to rotate dipole from 90° to θ is stored as PE. $dW = \tau d\theta = mB \sin \theta d\theta$. Integrating from 90° to θ , $U = -mB \cos \theta = -\vec{m} \cdot \vec{B}$. **Q37:** (a) Hysteresis loss (energy dissipated per cycle per unit volume). (b) Material P (narrow curve) is better for transformers due to low hysteresis loss, preventing excessive heating. **Q38:** Dia: lines avoid the material. Para: lines pass preferentially through it. Distinguishing property: Magnetic Permeability ($\mu_r < 1$ for dia, $\mu_r > 1$ for para). **Q39:** Retentivity: Magnetization left when H becomes zero. Coercivity: Reverse H needed to

demagnetize completely. Soft iron has higher retentivity; Steel has higher coercivity. **Q40:** $\tau = mB \sin \theta \implies 4.5 \times 10^{-2} = m \times 0.25 \times \sin 30^\circ \implies m = 0.36 \text{ A} \cdot \text{m}^2$. Stable equilibrium ($\theta = 0^\circ$): $U = -mB \cos 0^\circ = -0.36 \times 0.25 = -0.09 \text{ J}$. **Q41:** (i) Axial: $B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}$. (ii) Equatorial: $B = \frac{\mu_0}{4\pi} \frac{m}{r^3}$. Both depend on $1/r^3$. **Q42:** Ferromagnetic (since $\mu_r \gg 1$). Heated beyond critical temperature, it loses ferromagnetism and becomes paramagnetic. This is called the Curie Temperature. **Q43:** (a) Zero. (b) Domains parallel to the field grow in size, and other domains rotate to align with the field, causing strong magnetization. (c) $\mu_r = 1 + \chi$. (d) Thermal agitation in paramagnets opposes alignment of permanent dipoles, reducing χ as T increases. Diamagnetism relies on orbital distortion, largely unaffected by thermal motion. **Q44:** (a) Heat energy dissipated per unit volume per cycle of magnetization. (b) High retentivity (strong magnet) and high coercivity (resistant to demagnetization). (c) Low hysteresis loss, high permeability, and low coercivity. (d) Retentivity: retained magnetism at $H = 0$. Coercivity: reverse H to make $M = 0$. **Q45:** (A) $\vec{\tau} = \vec{m} \times \vec{B}$. (B) $m = NIA = 20 \times 12 \times 0.01 = 2.4 \text{ A} \cdot \text{m}^2$. $\tau = mB \sin 30^\circ = 2.4 \times 0.80 \times 0.5 = 0.96 \text{ N} \cdot \text{m}$. (C) (i) $\theta = 0^\circ$ (stable), (ii) $\theta = 180^\circ$ (unstable). **Q46:** (A) Para: atoms have permanent net magnetic moment. Dia: paired electrons cancel moments; external field induces opposite moments. (B) Ferro has microscopic domain structures that align cooperatively, creating huge magnetization compared to weak independent atomic alignment in para. (C) Paramagnetic (positive and small). $\mu_r = 1 + \chi = 1.000026 > 1$. **Q47:** (A) $\oint \vec{B} \cdot d\vec{S} = 0$, while electrostatics is $\oint \vec{E} \cdot d\vec{S} = q/\epsilon_0$. Reason: Magnetic charges (monopoles) do not exist isolated. (B) (i) $H = nI = 1000 \times 2 = 2000 \text{ A/m}$. (ii) $B = \mu_0 \mu_r H = 4\pi \times 10^{-7} \times 400 \times 2000 \approx 1.005 \text{ T}$. (iii) $M = (\mu_r - 1)H = 399 \times 2000 = 7.98 \times 10^5 \text{ A/m}$. **Q48:** (A) It will experience both a net force (moving it toward the stronger or weaker field) and a torque (aligning it). (B) (i) $B_{axis} = \frac{\mu_0}{4\pi} \frac{2m}{r^3} = 10^{-7} \times \frac{2(0.48)}{0.1^3} = 0.96 \times 10^{-4} \text{ T}$ along $\vec{m}$. (ii) $B_{eq} = \frac{\mu_0}{4\pi} \frac{m}{r^3} = 0.48 \times 10^{-4} \text{ T}$ opposite to $\vec{m}$. (C) Draw standard continuous loops emerging from N and entering S. **Q49:** (a) Soft iron. (b) The total magnetic field is vastly increased ($B = \mu_r B_0$) because the highly permeable core gets magnetized. (c) Increase the current (I) and increase the number of turns per unit length (n). (d) No, steel has high retentivity; it would remain permanently magnetized even after the current is switched off. **Q50:** (A) Draw broad loop for steel, narrow tall loop for soft iron. (B) Soft iron: higher retentivity, lower coercivity, lower hysteresis loss. Steel: lower retentivity, higher coercivity, higher hysteresis loss. (C) (i) Steel (high coercivity prevents demagnetization). (ii) Soft iron (low coercivity and narrow loop minimizes heat loss in AC cycles).